

JPE Replication Report

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1 Paper Identification

ID	20241434
Author	Ganapati
Title	From Heterogeneous Firms to Heterogeneous Trade Elasticities: The Aggregate Implications of Firm Export Decisions
Iteration	1

Some useful information up front:

- [JPE Data and Code Availability Policy](#)
- [Step by step guidance from Data Editor for Authors](#)
- [Template README](#)

2 Summary

In assessing compliance with our [Data and Code Availability Policy](#), our reproducibility team has identified the following issues, which we ask you to address:

1. We flag by default all instances of Potential Personal Identifiable Information (PII). Please verify that you have the right to publish those - if applicable.
2. There is a spelling issue in your stata code with causes problems on case-sensitive systems like ours (Ubuntu)
3. We hit a bug where a variable seems not get created as intended and we had to stop there. Please advise.

3 General

- ☒ Data Availability and Provenance Statements
 - ☒ Statement about Rights - Part 1: Right to use the data (e.g. *did the authors lawfully obtain the data*)
 - ☒ Statement about Rights - Part 2: Right to publish the data (e.g. *are human subjects involved and did permission been granted to publish their data?*)
 - ☐ License for Data (optional, but recommended)
 - ☒ Details on each Data Source
- ☒ Dataset list
- ☒ Computational requirements
 - ☒ Software Requirements
 - ☒ Controlled Randomness (as necessary)
 - ☒ Memory, Runtime, and Storage Requirements
- ☒ Description of programs/code
 - ☐ License for Code (Optional, but recommended)
- ☐ {REPLICATOR} to Replicators
 - ☐ Details (as necessary)
- ☒ List of tables and programs
- ☒ References (Optional, but recommended)

4 High-level Description of Research Project

This research project...

- ☒ uses observational data.
- ☒ uses *confidential* observational data.
- ☐ uses data generated via experiment or trial by the authors (in field or lab).
 - ☐ For lab experiments: The code to implement the experimental protocol (z-Tree, o-Tree etc) is included in the package.
 - ☐ A PDF copy of IRB approval is contained in the package.
 - ☐ A PDF document outlining the design of the experiment, including information on the selection and eligibility of participants is included.
 - ☐ A PDF copy of the instructions given to participants in both original language and an English translation is included.
- ☐ uses data generated via survey by the authors (in field or online)
 - ☐ The programs used to administer the survey (e.g. qualtrics survey file) are included.
 - ☐ A PDF copy of IRB approval is contained in the package.

- ☐ A PDF of the original questionnaire(s) and information on the selection and eligibility of participants is included.
- ☐ is a simulation study: the data are generated by an algorithm.
- ☐ is a numerical illustration: no data are used but code is used to produce illustrative output (tables, figures)

5 Data description

5.1 Data Sources

Data Source / file name	public	private	DOI/URL	Access described	cited in README	cited in paper
UNCTAD TRAINS – Tariff Data	no	no	yes	yes	yes	yes
World Bank EDD – Exporter Dynamics Database	yes	no	yes	yes	yes	yes
World Bank EDD – Colombian Microdata	no	yes	no	yes	yes	yes
CEPII – Gravity Variables	yes	no	yes	yes	yes	yes
WIOD – World Input-Output Database (2016)	yes	no	yes	yes	yes	yes
Teti Tariffs – Feodora Teti Data	yes	no	yes	yes	yes	yes
OECD SDBS/TEC – Firm Demographics & TEC	yes	no	yes	yes	yes	yes
US Census – Manufacturer Enterprises (2012)	yes	no	yes	yes	yes	yes
Australia Data – Characteristics of Exporters	yes	no	yes	yes	yes	yes
China Statistical Yearbook (2019)	yes	no	yes	yes	yes	yes
China Export Data	no	yes	no	yes	yes	yes
World Bank Enterprise Surveys	yes	no	yes	yes	yes	yes
GSP Rates – Constructed from TRAINS	no	no	no	yes	yes	yes
Eora MRIO – Multi-Region Input-Output Data	yes	no	yes	yes	yes	yes

6 Requirements

- ☒ README is in TXT, MD, or PDF format
- ☒ README is accessible at the root of the package (not inside a folder)
- ☐ Title conforms to guidance (starts with “Data and Code for: Title of Paper” or “Code for: Title of Paper”, is properly capitalized)
- ☒ Authors (with affiliations) are listed in the same order as on the paper

{REQUIRED} Please review the title of the deposit as per our guidelines.

6.1 Stated Requirements

- ☐ No requirements specified
- ☒ Software Requirements specified as follows:

Stata 16.1/MP

Matlab 2023a
- ☒ Computational Requirements specified as follows:

OS X environment (M2 processor with 16 cores and 32 GB) and a 1 TB SSD for all the programs except ‘run_mc_master’ which was run on an external Linux cluster with 100 available cores (32 GB addressable each). Programs do require z-shell scripting, so they do need an *nix/OSX environment.
- ☒ Time Requirements specified as follows:

8-24 hours
- ☒ Requirements are complete.

7 Analyzing Data and Code Files

7.1 Potential Personal Identifiable Information (PII)

⚠ We found the following instances of potentially personally identifying information. This may be completely legitimate but might be worth checking. As a reminder, privacy legislation in many countries (e.g. GDPR in EU) prohibits the dissemination of personal identifiable information without prior (and documented) consent of individuals. If indeed you want to publish such information with your replication package, you should probably have obtained IRB approval for this - please check!

Summary: - Data files with PII indicators: 52 - Variables flagged in data: 169 - Code files with PII references: 76 - PII references in code: 1216

7.1.1 Summary of Flagged Files

File Type	File	Variables/References	PII Categories
Data	AUS_cleaned.csv	1	country
Data	COL_EXP_2012.dta	2	country
Data	CYD_manuf.dta	1	country
Data	CYH2D_manuf.dta	2	country
Data	CYH2D_manuf_1000usd.dta	2	country
Data	CY_manuf.dta	1	country
Data	Combined_Industrial_Enterprises_Indicators.xlsx	1	son
Data	Countries_V202102.dta	4	country, name, lon
Data	EDDVariableNames.xlsx	1	name
Data	Eora26_2012_bp.zip	1	block, loc
Data	Final_Data.dta	1	country, name
Data	Full_Trade.csv	1	lat
Data	Gravity_V202102.dta	19	country, lat, lon
Data	SDBS_BDI_ISIC4_11042019204932050.csv	2	loc, location, country
Data	SSIS_BSC_ISIC4_11042019175351230.csv	2	loc, location, country
Data	Simulated_data_LogNormal_alt.mat	2	lon
Data	TES3_04052018220827756.csv	2	country
Data	WIOT2000_October16_ROW.dta	1	country
Data	WIOT2001_October16_ROW.dta	1	country
Data	WIOT2002_October16_ROW.dta	1	country
Data	WIOT2003_October16_ROW.dta	1	country
Data	WIOT2004_October16_ROW.dta	1	country
Data	WIOT2005_October16_ROW.dta	1	country
Data	WIOT2006_October16_ROW.dta	1	country
Data	WIOT2007_October16_ROW.dta	1	country
Data	WIOT2008_October16_ROW.dta	1	country
Data	WIOT2009_October16_ROW.dta	1	country
Data	WIOT2010_October16_ROW.dta	1	country
Data	WIOT2011_October16_ROW.dta	1	country
Data	WIOT2012_October16_ROW.dta	1	country
Data	WIOT2013_October16_ROW.dta	1	country
Data	WIOT2014_October16_ROW.dta	1	country
Data	aff_download_readme_ann.txt	2	zip, lat

File Type	File	Variables/References	PII Categories
Data	country_codes_V202201.csv	3	country, name
Data	dist_cepii.dta	7	lat, lon, country
Data	dist_cepii.zip	1	lon
Data	dist_cepii_bra.dta	7	lat, lon, country
Data	exports_2007.dta	1	country
Data	exports_2008.dta	1	country
Data	exports_2009.dta	1	country
Data	exports_2010.dta	1	country
Data	exports_2011.dta	1	country
Data	exports_2012.dta	1	country
Data	fpart_concordance.dta	1	country
Data	geo_cepii.dta	25	country, name, loc, city, lat, degree, lon, son
Data	geo_cepii.xls	15	country, loc, city, lat, lon
Data	geo_cepii_maincity.dta	25	country, name, loc, city, lat, degree, lon, son
Data	gravdata_cepii.dta	13	lat, lon
Data	index_readme.txt	1	block, loc
Data	tariff2010_beta1-2024-12.zip	1	name
Data	tariff2012_beta1-2024-12.zip	1	name
Data	tariff2014_beta1-2024-12.zip	1	name
Code	AppendixFigure_LogPareto.m	5	lon, loc, location, lat
Code	CF_agg_plot.m	1	lat
Code	CF_diagnostic.m	3	lat, loc, location
Code	CF_flow_plot.m	2	loc, location, lat
Code	Figure_Sequencer.m	5	city, lon
Code	GFT.m	7	lat, lon
Code	GFT.m	7	lat, lon
Code	GFTG_v10.m	4	lon
Code	GFTG_v10.m	4	lon
Code	GFT_Script_v17_EKK.m	19	lon, lat, city, loc, location
Code	GFT_agg_plot.m	2	lat, loc, location
Code	GFT_division.m	1	city, loc, location
Code	GFT_division2.m	1	city, loc, location
Code	GFT_fct_general_fast.m	27	lon, lat, city
Code	GFT_pre.m	4	lat, lon
Code	GFT_pre1.m	4	lat, lon
Code	GFT_script_v17.m	139	lon, city, lat, name, loc, location

File Type	File	Variables/References	PII Categories
Code	GFT_v9.m	5	lon
Code	GFT_v9.m	5	lon
Code	GMM_gravity.m	3	lon, second
Code	GMM_gravity.m	5	lon, second
Code	GMM_gravity_log_normal.m	6	lon, lat, second
Code	GMM_wrapper_gravity.m	20	lon, dob, second
Code	GMM_wrapper_gravity.m	21	lon, name, dob, second
Code	GenerateBaseline.m	7	name, lon, lat
Code	Graphs_GSP.m	21	city, name, lat, loc, location
Code	Graphs_TC.m	33	city, name, lat, loc, location
Code	LinearCF.m	23	second, name, lon, city, lat
Code	LinearCF_wrapper_v3.m	1	lat
Code	MasterProgram.do	7	name, loc, lat
Code	Master_script_gravity_v3.m	94	lon, name, city, lat, loc, location
Code	QuantilesBaseline.m	1	lat
Code	QuantilesColombia.m	1	dob
Code	Quantiles_Colombia_Wrapper.m	10	name, country
Code	Script_GMM_simulation.m	28	lat, lon, loc, name, second
Code	Script_QQ_simulation.m	27	lat, lon, loc, name, second, city
Code	Simulation_Data_v10.m	51	lat, city, lon, loc, location, name
Code	Test_Code.m	2	lon
Code	create_elasticity_functions.m	19	city, lon, lat, son
Code	figure_plotter_QQ_minimal.m	9	lat, degree, name
Code	figures_gen.m	41	city, lat, son, name, loc, location, child, dob
Code	function_interp_wrapper.m	9	lon, city, lat, son
Code	function_interp_wrapper_GMM.m	8	lon, city
Code	generate_QQ_plot.m	19	lat
Code	generate_eps.m	1	lon
Code	generate_pdf_entry.m	5	lat, name, loc, location
Code	get_fct_form_params.m	4	city
Code	graph_overlay.m	31	lon, lat, loc, location, city, name
Code	hs_routine_gravity.m	8	lon, city
Code	inv_unique_interp.m	1	lat
Code	logpareto_inv_shifted.m	3	loc, location, lat
Code	make_data_combo.m	19	lon, name, lat, son

File Type	File	Variables/References	PII Categories
Code	make_data_combo_hs_multi.m	15	lon, name, lat, son
Code	make_data_simulation.m	3	lat, name, son
Code	make_elasticity_combo.m	3	city, lon
Code	p01_TRAINS_clean_data_v1_AAG.do	52	country, name, zip, loc, lat, lon
Code	p02_ChinaData_Morrow.do	3	loc, country, name
Code	p02_eora_trade_matrix_clean.do	9	name, lon
Code	p02_quantiles_colombia_firmexports.do	1	name
Code	p03_Create_Tariffs.do	6	name
Code	p04_Sample_Creation.do	130	name, country, lon, loc, lat, location, census, sex
Code	p05_Sample_Creation_HS2.do	9	name, lon
Code	p06_ReducedForm.do	7	lon, loc
Code	p_overlay_elasticity.m	101	city, name, lon, lat, loc, location
Code	plot_Er_vs_log_n.m	3	name, lon, loc, location
Code	plot_logcorr_pareto.m	1	lon
Code	plot_theta_c_vs_log_n.m	2	name, loc, location
Code	plot_theta_combined.m	6	lon, name, loc, location
Code	plot_theta_e_vs_log_n.m	9	lon, lat, name, loc, location
Code	plot_theta_i_vs_log_n.m	6	lon, name, loc, location
Code	recover_elasticities_LN.m	6	lon, lat
Code	regQQ.m	4	lon, lat
Code	run_mc_master.m	34	lat, city, lon, name
Code	solve_FP.m	19	lat, city, lon, country, name
Code	vline.m	2	loc, location, lat
Code	vline.m	2	loc, location, lat

See [Appendix](#) for detailed listing of all flagged instances.

7.1.2 File Paths Report

Generated on 2026-03-10T18:08:47.989

Warning: Our search on file path types is imperfect and incurs both type 1 and type 2 errors. We aim to strike a reasonable balance between both. The below table is therefore only indicative. Detailed listings can be found in the appendix to this report.

In this table we analyse all files which contain source code (not data and documentation), and we report the grand total of each type of filepath we encountered.

Please check and replace any `windows` filepaths with unix compliant paths, insofar as this is possible in your setup. (Notice that `STATA` allows `/` to be a filepath separator on a windows platform, hence this is a requirement for *all* `STATA` applications.)

Files Analyzed	Windows Paths	Unix Paths	Mixed Paths	Drive Letters (<code>C:\</code> etc)
196	2	30	7	0

7.1.3 Duplicate Files Report

We found the following duplicate files:

Filename	Size (MB)	Checksum (MD5)
__MACOSX/Code/Appendix_LogPareto/._DS_Store	0.0	0bed7e90c2bade9763fa18f1fb44
__MACOSX/Code/ColombiaTests/._DS_Store	0.0	0bed7e90c2bade9763fa18f1fb44
__MACOSX/Code/GMM_estimation_v3/._DS_Store	0.0	0bed7e90c2bade9763fa18f1fb44
__MACOSX/Code/MonteCarlo/._DS_Store	0.0	0bed7e90c2bade9763fa18f1fb44
__MACOSX/Code/MonteCarlo/output/._DS_Store	0.0	0bed7e90c2bade9763fa18f1fb44
__MACOSX/Code/MonteCarlo/output/plots/._DS_Store	0.0	0bed7e90c2bade9763fa18f1fb44
__MACOSX/Crosswalks/._DS_Store	0.0	0bed7e90c2bade9763fa18f1fb44
__MACOSX/Data/._DS_Store	0.0	0bed7e90c2bade9763fa18f1fb44
__MACOSX/Data/Australia/._DS_Store	0.0	0bed7e90c2bade9763fa18f1fb44
__MACOSX/Data/CEPII/._DS_Store	0.0	0bed7e90c2bade9763fa18f1fb44
__MACOSX/Data/CEPII/BACI_HS92_V202201/._DS_Store	0.0	0bed7e90c2bade9763fa18f1fb44
__MACOSX/Data/CEPII/Gravity/._DS_Store	0.0	11050c2745c99ec8dacff161c2a0
__MACOSX/Data/China_Exports/._DS_Store	0.0	0bed7e90c2bade9763fa18f1fb44
__MACOSX/Data/China_Exports/Final/._DS_Store	0.0	0bed7e90c2bade9763fa18f1fb44
__MACOSX/Data/China_Exports/Raw/._DS_Store	0.0	0bed7e90c2bade9763fa18f1fb44
__MACOSX/Data/China_Statistical_Yearbook/._DS_Store	0.0	0bed7e90c2bade9763fa18f1fb44
__MACOSX/Data/China_Statistical_Yearbook/Raw/._DS_Store	0.0	0bed7e90c2bade9763fa18f1fb44
__MACOSX/Data/China_Statistical_Yearbook/Raw/China Statistical Yearbook 2019_files/._DS_Store	0.0	0bed7e90c2bade9763fa18f1fb44
__MACOSX/Data/EDD_Data/._DS_Store	0.0	0bed7e90c2bade9763fa18f1fb44
__MACOSX/Data/EDD_Data/CYD_manuf/._DS_Store	0.0	0bed7e90c2bade9763fa18f1fb44
__MACOSX/Data/EDD_Data/CY_manuf/._DS_Store	0.0	0bed7e90c2bade9763fa18f1fb44
__MACOSX/Data/EDD_Data/Dictionary/._DS_Store	0.0	0bed7e90c2bade9763fa18f1fb44
__MACOSX/Data/EORA/._DS_Store	0.0	0bed7e90c2bade9763fa18f1fb44
__MACOSX/Data/EORA/Clean_data/._DS_Store	0.0	0bed7e90c2bade9763fa18f1fb44
__MACOSX/Data/EORA/Output/._DS_Store	0.0	0bed7e90c2bade9763fa18f1fb44
__MACOSX/Data/EORA/Raw_data/._DS_Store	0.0	0bed7e90c2bade9763fa18f1fb44
__MACOSX/Data/EORA/Raw_data/Eora26_2012_bp/._DS_Store	0.0	0bed7e90c2bade9763fa18f1fb44
__MACOSX/Data/GSP/._DS_Store	0.0	0bed7e90c2bade9763fa18f1fb44
__MACOSX/Data/Int/._DS_Store	0.0	0bed7e90c2bade9763fa18f1fb44
__MACOSX/Data/OECD/._DS_Store	0.0	0bed7e90c2bade9763fa18f1fb44
__MACOSX/Data/OECD/SDBS Business Demography Indicators/._DS_Store	0.0	0bed7e90c2bade9763fa18f1fb44
__MACOSX/Data/TRAINS/._DS_Store	0.0	0bed7e90c2bade9763fa18f1fb44
__MACOSX/Data/TRAINS/cleaned_data/._DS_Store	0.0	0bed7e90c2bade9763fa18f1fb44
__MACOSX/Data/TRAINS/collapsed/._DS_Store	0.0	0bed7e90c2bade9763fa18f1fb44
__MACOSX/Data/Teti/._DS_Store	0.0	0bed7e90c2bade9763fa18f1fb44
__MACOSX/Data/US_Census/._DS_Store	0.0	11050c2745c99ec8dacff161c2a0
__MACOSX/Data/US_Census/2012_US_Mfg/._DS_Store	0.0	0bed7e90c2bade9763fa18f1fb44

Filename	Size (MB)	Checksum (MD5)
__MACOSX/Data/WB_Enterprise_Data/._DS_Store	0.0	0bed7e90c2bade9763fa18f1fb44
__MACOSX/Data/WIOD/._DS_Store	0.0	0bed7e90c2bade9763fa18f1fb44
__MACOSX/Data/WorldBankData/._DS_Store	0.0	0bed7e90c2bade9763fa18f1fb44
__MACOSX/Data/WorldBankData/Clean data/._DS_Store	0.0	0bed7e90c2bade9763fa18f1fb44
Code/MonteCarlo/functions/evknots_make_deriv.m	0.0	47df39afafbdb1503a71b5fd59bcc
Code/MonteCarlo/functions/evknots_make.m	0.0	31974cba393db94eecbcd9a5cee
Code/MonteCarlo/functions/GFT_v2/GFTG_v10.m	0.0	bbfe609423a0fbc935db72314e
Code/MonteCarlo/functions/GFT_v2/GFT_v9.m	0.0	3e91184a622cf9c9026d6989aae
Code/MonteCarlo/functions/GFT_v2/GFT.m	0.0	6f8380bcb7fd0162b0bfaa16144e
Code/MonteCarlo/functions/vline.m	0.0	91f059f68599452e21a3a7fdb83f
Data/CEPII/BACI_HS92_V202201/.DS_Store	0.01	df2fbeb1400acda0909a32c1cf6b
Data/China_Exports/Final/.DS_Store	0.01	df2fbeb1400acda0909a32c1cf6b
Data/China_Exports/Raw/.DS_Store	0.01	df2fbeb1400acda0909a32c1cf6b
Data/EDD_Data/CYD_manuf/.DS_Store	0.01	df2fbeb1400acda0909a32c1cf6b
Data/EDD_Data/CY_manuf/.DS_Store	0.01	df2fbeb1400acda0909a32c1cf6b
Data/EORA/Clean_data/.DS_Store	0.01	df2fbeb1400acda0909a32c1cf6b
Data/EORA/Output/.DS_Store	0.01	df2fbeb1400acda0909a32c1cf6b
Data/OECD/SDBS Business Demography Indicators/.DS_Store	0.01	df2fbeb1400acda0909a32c1cf6b
Data/TRAINS/cleaned_data/.DS_Store	0.01	df2fbeb1400acda0909a32c1cf6b
Data/TRAINS/collapsed/.DS_Store	0.01	df2fbeb1400acda0909a32c1cf6b
Data/WorldBankData/Clean data/.DS_Store	0.01	df2fbeb1400acda0909a32c1cf6b
replication.md	0.03	ddd0191409b40fc2b19fac751664
Data/EORA/Raw_data/Eora26_2012_bp/import/miranda/Eoras/Phase199/Loop082/simplified/Eora26_2012_bp_VA.txt	0.37	d523cd2a7499f104e3e6078cdd3
Data/China_Statistical_Yearbook/Raw/China Statistical Yearbook 2019_files/E1302_002.jpg	0.61	6a04329e0426be7693de12c261f
Sharat G - AAG_2026_03_06.pdf	11.1	b33a0577f08794dcd4302cf8fee
Data/EORA/Raw_data/Eora26_2012_bp/import/miranda/Eoras/Phase199/Loop082/simplified/Eora26_2012_bp_FD.txt	68.85	7d15c8bc188563ed26adc8ef677
Data/EORA/Raw_data/Eora26_2012_bp/import/miranda/Eoras/Phase199/Loop082/simplified/Eora26_2012_bp_T.txt	296.84	c1b90cf89b5fc6b223d10c38363

7.1.4 Zero Size Files Report

We did not find any zero sized files.

7.1.5 Large Files Report

We found the following files larger than 100MB:

Filename	Size (MB)
Data/TRAINS/raw_data/TRAINS1995.zip	149.19
Data/EORA/Raw_data/Eora26_2012_bp/Eora26_2012_bp_Q.txt	164.76
Data/TRAINS/raw_data/TRAINS1996.zip	211.02
Data/TRAINS/raw_data/TRAINS1998.zip	214.97

Filename	Size (MB)
Data/TRAINS/raw_data/TRAINS1997.zip	233.99
Data/EORA/Raw_data/Eora26_2012_bp.zip	245.51
Data/TRAINS/raw_data/TRAINS1999.zip	252.15
Data/EORA/Raw_data/Eora26_2012_bp/Eora26_2012_bp_T.txt	296.84
Data/EORA/Raw_data/Eora26_2012_bp/import/miranda/Eoras/Phase199/Loop082/simplified/Eora26_2012_bp_T.txt	296.84
Data/TRAINS/raw_data/TRAINS2000.zip	327.46
Data/CEPII/BACI_HS92_V202201/BACI_HS92_Y2000_V202201.csv	354.76
Data/CEPII/BACI_HS92_V202201/BACI_HS92_Y2001_V202201.csv	366.9
Data/TRAINS/raw_data/TRAINS2001.zip	371.3
Data/CEPII/BACI_HS92_V202201/BACI_HS92_Y2002_V202201.csv	376.96
Data/CEPII/BACI_HS92_V202201/BACI_HS92_Y2003_V202201.csv	389.46
Data/TRAINS/raw_data/TRAINS2003.zip	391.18
Data/TRAINS/raw_data/TRAINS2002.zip	396.85
Data/CEPII/BACI_HS92_V202201/BACI_HS92_Y2004_V202201.csv	407.54
Data/TRAINS/raw_data/TRAINS2004.zip	409.83
Data/CEPII/BACI_HS92_V202201/BACI_HS92_Y2005_V202201.csv	422.08
Data/CEPII/BACI_HS92_V202201/BACI_HS92_Y2006_V202201.csv	438.47
Data/CEPII/BACI_HS92_V202201/BACI_HS92_Y2007_V202201.csv	449.43
Data/CEPII/BACI_HS92_V202201/BACI_HS92_Y2009_V202201.csv	449.46
Data/CEPII/BACI_HS92_V202201/BACI_HS92_Y2008_V202201.csv	455.21
Data/TRAINS/raw_data/TRAINS2005.zip	457.02
Data/CEPII/BACI_HS92_V202201/BACI_HS92_Y2010_V202201.csv	462.52
Data/CEPII/BACI_HS92_V202201/BACI_HS92_Y2011_V202201.csv	470.89
Data/TRAINS/raw_data/TRAINS2006.zip	475.98
Data/TRAINS/raw_data/TRAINS2014.zip	483.25
Data/CEPII/BACI_HS92_V202201/BACI_HS92_Y2012_V202201.csv	484.09
Data/TRAINS/raw_data/TRAINS2007.zip	493.25
Data/CEPII/BACI_HS92_V202201/BACI_HS92_Y2013_V202201.csv	493.35
Data/CEPII/BACI_HS92_V202201/BACI_HS92_Y2014_V202201.csv	495.26
Data/TRAINS/raw_data/TRAINS2010.zip	495.47
Data/TRAINS/raw_data/TRAINS2017.zip	500.14
Data/TRAINS/raw_data/TRAINS2009.zip	505.4
Data/TRAINS/raw_data/TRAINS2012.zip	508.66
Data/TRAINS/raw_data/TRAINS2018.zip	510.35
Data/TRAINS/raw_data/TRAINS2011.zip	522.01
Data/TRAINS/raw_data/TRAINS2008.zip	523.42
Data/TRAINS/raw_data/TRAINS2013.zip	528.29
Data/TRAINS/raw_data/TRAINS2015.zip	538.0

Filename	Size (MB)
Data/TRAINS/raw_data/TRAINS2016.zip	542.54
Data/CEPII/Gravity/gravdata_cepii.dta	581.64
Data/Teti/tariff2012_beta1-2024-12.zip	593.5
Data/Teti/tariff2010_beta1-2024-12.zip	593.55
Data/Teti/tariff2014_beta1-2024-12.zip	599.49
Data/EDD_Data/CYH2D_manuf/CYH2D_manuf_1000usd.dta	810.31
Data/EDD_Data/CYH2D_manuf/CYH2D_manuf.dta	814.36
Data/CEPII/Gravity_dta_V202102/Gravity_V202102.dta	1183.59

7.2 Code description

- There are eight Stata do-files, six Matlab .m files and one master do-file (Stata and Matlab combined).

cloc		github.com/AIDanial/cloc v 2.06 T=0.87 s (143.4 files/s, 565528.2 lines/s)		
Language	files	blank	comment	code
CSV	11	0	0	460000
Text	13	104	0	16765
MATLAB	83	1822	1769	6627
HTML	8	132	61	2612
Stata	9	599	274	2048
Markdown	1	44	0	132
---	---	---	---	---
SUM:	125	2701	2104	488184

- ☒ The replication package contains a "main" or "master" file(s) which calls all other auxiliary programs.

NOTE: In-text numbers that reference numbers in tables do not need to be listed. Only in-text numbers that correspond to no table or figure need to be listed.

7.3 Computing Environment of the Replicator

- Nuvolos cloud
- Stata/MP 16.0
- Matlab R2023a

8 Replication

8.1 Stata

- Downloaded code and data from dropbox.
- unzipped all the necessary datafiles.
- Ran the master do-file as per README, in Stata. Master do-file did not run smoothly in one go, I had to make some minor changes:

8.1.1 p02_qunatiles_colombia_firmexports.do

- the ROOT in line 5 said "DATA" instead of "Data", as a result datasets in the code could not be found. Changed it accordingly to "Data".

8.1.2 p04_Sample_Creation.do

- multiple ROOT's in the code said "DATA" instead of "Data" and could therefore not call the right datasets. Changed all of them it accordingly to "Data".
- The script fails at the point where it attempts to access the variable `SurvivalRate1_average`. After reviewing the code, the issue appears to originate in the block between lines 584 and 632. In this section, the average survival-rate variables (`SurvivalRate1_average`, `SurvivalRate2_average`, `SurvivalRate3_average`) are apparently not created. The underlying variables (`SurvivalRate1`, `SurvivalRate2`, `SurvivalRate3`) are empty at that stage of the script. Running `count` after the filtering steps states zero observations. The SDBS datasets itself appears to be complete, but the combination of filters (`Variable`, `SEC == "10_33"`, `SizeClass == "Total"`, `TIME == year`) does maybe remove all observations in this part of the code. Because the dataset is empty, `egen mean()` cannot compute an average. Due to the use of `capture`, the resulting error is likely suppressed, and the average variables are not created. Later in the script, the variables can not be found and the code breaks down.

These survival-rate datasets are required for subsequent steps in the replication. I tried running other parts of the code but naturally it does not work without all the necessary datasets. Is the filtering currently correct or is there something else that I am missing?

```

614 insheet using "$ROOT/Data/OECD/SDBS Business Demography
615 Indicators/SDBS_BDI_ISIC4_11042019204932050.csv", clear names case
616 keep if Variable == "1-year survival rate"
617 keep if SEC == "10_33" // Keep MFG
618 keep if SizeClass == "Total"
619 keep if TIME == ${year}
620 rename LOCATION iso3_o
621 rename Value SurvivalRate1
622 label var SurvivalRate1 "OECD 1-year survival rate"
623 keep iso3_o SurvivalRate1
624 drop if SurvivalRate1 >= 100
625 merge 1:1 iso3_o using "$ROOT/Data/Int/SurvivalRate2${year}", nogen
626 merge 1:1 iso3_o using "$ROOT/Data/Int/SurvivalRate3${year}", nogen
627 capture egen SurvivalRate1_average = mean(SurvivalRate1)
628 capture label var SurvivalRate1_average "Average OECD 1-year survival rate"
629 capture egen SurvivalRate2_average = mean(SurvivalRate2)
630 capture label var SurvivalRate2_average "Average OECD 2-year survival rate"
631 capture egen SurvivalRate3_average = mean(SurvivalRate3)
632 capture label var SurvivalRate3_average "Average OECD 5-year survival rate"
633 saveold "$ROOT/Data/Int/SurvivalRate${year}", replace
634 end
635

```

snapshot

9 Findings

9.1 Missing Requirements

- ☒ Software Requirements
 - ☒ Stata
 - 'Require' package (necessary for reghdfe)

[REQUIRED] Please amend README to contain complete requirements.

You can copy the section above, amended if necessary.

9.2 Data Preparation Code

- Code/p01_TRAINS_clean_data_v1_AAG.do ran without any error.
- Code/p02_quantiles_colombia_firmexports.do ran without any error.
- Code/p02_eora_trade_matrix_clean.do ran without any error.
- Code/p02_ChinaData_Morrow.do ran without any error.
- Code/p03_Create_Tariffs.do ran without any error.
- Code/p04_Sample_Creation.do broke down with described error, script is not able to prepare the necessary datasets.
- Code/p05_Sample_Creation_HS2.do could not run script without the necessary datasets.
- Code/p06_ReducedForm.do could not run script without the necessary datasets.
- Not able to run any of the Matlab code without completing the data construction part.

9.3 Tables and Figures

Figure/Table #	Program	Reproduced?
Figure 1	Matlab	No
Figure 2	Matlab	No
Figure 3	Matlab	No
Figure 4	Matlab	No
Figure 5	Matlab	No
Figure 6	Matlab	No
Table 1	Matlab	No
Figure 0A.1	Matlab	No
Figure 0A.2	Matlab	No
Figure 0A.3	Matlab	No
Figure 0A.4	Matlab	No
Figure 0A.5	Matlab	No
Figure 0A.6	Matlab	No
Table 0A.1	Matlab	No
Table 0A.2	Matlab	No
Table 0A.3	Matlab	No
Figure 0A.7	Matlab	No
Figure 0A.8	Matlab	No
Figure 0A.9	Matlab	No
Figure 0A.10	Matlab	No
Figure 0A.11	Matlab	No
Figure 0A.12	Matlab	No
Figure 0A.13	Matlab	No
Figure 0A.14	Matlab	No
Figure 0A.15	Matlab	No
Figure 0A.16	Matlab	No
Figure 0A.17	Matlab	No
Figure 0A.18	Matlab	No
Figure 0A.19	Matlab	No
Figure 0A.20	Matlab	No
Figure 0A.21	Matlab	No
Figure 0A.22	Matlab	No
Figure 0A.23	Matlab	No
Figure 0A.24	Matlab	No
Figure 0A.25	Matlab	No
Figure 0A.26	Matlab	No
Figure 0A.27	Matlab	No
Figure 0A.28	Matlab	No

Figure/Table #	Program	Reproduced?
Figure 0A.29	Matlab	No
Figure 0A.30	Matlab	No
Figure 0A.31	Matlab	No
Figure 0A.32	Matlab	No
Figure 0A.33	Matlab	No

I could not reproduce any results, so no comparisons with the paper’s figures or tables was possible.

9.4 Other Issues

I had some additional comments on the README:

- the README states a required storage space ranging from 25GB to 250GB. I reached my full storage capacity for this replication (250GB) and had to delete some of the raw TARIFF datasources before I could continue with the replication. This might be something to explain further in the README, e.g. is there a certain point in the replication in which some raw datasources can be deleted in case a replicator works with limited data capacity?
- I am currently missing an overview in the README of all the final datasets that are generated by specific scripts and eventually used in the replication.
- A ‘for replicators’ section in the README, just a short section on how to run the code with clear instructions. Thereby making a distinction between running the master do-file (Stata & Matlab) or if someone prefers (or is not able to combine Stata and Matlab) running the scripts seperately.

10 Classification

As stated earlier in the report I was not able to complete the necessary scripts for the data preparation and was therefore not able to produce any output.

- ☐ full reproduction
- ☐ full reproduction with minor issues
- ☐ partial reproduction (see above)
- ☒ not able to reproduce most or all of the results (reasons see above)

10.1 Reason for incomplete reproducibility

- ☐ None.
- ☐ **Discrepancy in output** (either figures or numbers in tables or text differ)
- ☒ **Bugs in code** that were fixable by the replicator (but should be fixed in the final deposit)
- ☒ **Code missing**, in particular if it prevented the replicator from completing the reproducibility check

☒ **Data preparation code missing** should be checked if the code missing seems to be data preparation code
- ☐ **Code not functional** is more severe than a simple bug: it prevented the replicator from completing the reproducibility check
- ☐ **Software not available to replicator** may happen for a variety of reasons, but in particular (a) when the software is commercial, and the replicator does not have access to a licensed copy, or (b) the software is open-source, but a specific version required to conduct the reproducibility check is not available.
- ☐ **Insufficient time available to replicator** is applicable when (a) running the code would take weeks or more (b) running the code might take less time if sufficient compute resources were to be brought to bear, but no such resources can be accessed in a timely fashion (c) the replication package is very complex, and following all (manual and scripted) steps would take too long.
- ☐ **Data missing** is marked when data *should* be available, but was erroneously not provided, or is not accessible via the procedures described in the replication package
- ☐ **Data not available** is marked when data requires additional access steps, for instance purchase or application procedure.
- ☐ **Missing README** is marked if there is no README to guide the replicator, or the README is not in compliance with JPE requirements